

Randomised controlled short-term intervention pilot study on rye bran bread in prostate cancer.



The short-term effects of rye bran bread intake in prostate cancer were investigated. Ten men with conservatively treated prostate cancer were randomised to a daily supplement of 295 g of rye bran bread and eight men to 275 g of wheat bread (control) with similar fibre content for three weeks. Blood samples, ultrasound-guided core biopsies of the prostate, and urine samples were taken. In the rye group, there was a significant increase in plasma enterolactone, and the apoptotic index increased significantly from 2.1% (SD 1.3) to 5.9% (SD 1.8), $P < 0.005$ as measured by a TUNEL index in four cases in the rye group and seven cases in the control group. Besides a significant decrease in weight in both groups, only small changes were observed in plasma concentrations of prostate specific antigen (PSA), circulating sex hormones, excreted oestrogens, insulin-like growth factor (IGF)-I, and in the endothelial fibrinolytical system. High intake of rye bran bread is suggested to increase apoptosis in prostate tumours.